

## BEYOND THE SCORE

### Case 3 · Trace Sheet

*Design apparatus for BTS\_Case3\_Reading*

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#### 1 · CASE POSITION (HISTORICAL DESIGN)

Pilot 1, Case 3 of 4. Keying (as designed): Chinese-Malaysian intergenerational household, East Auckland, three-bedroom rental. Scenario: income-irregularity flag on a youth banking application; migrant family algorithmic exclusion. Trigger event: feature lockout citing ‘irregular household income patterns.’ Household: Aanya (17), Mei-Lin (46), Popo Jiahui (77); the father is an unvoiced presence working deliveries. Actor: institutional — automated risk profiling reaching the household through an app's withheld features; single-scene exposure. Lead card: the member the system addressed (Aanya). The open-banking mechanics modelled here were prospective at the time of the pilot; major-bank data-holder designation under the Customer and Product Data framework commenced 1 December 2025.

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#### 2 · CORPUS ANCHORS

Aanya: filial expectation and internalised tension — Wang (2018); Ip (2003); Ran & Liu (2023); Peiris-John et al. (2025). Model-minority pattern and youth mental health — Asian Family Services (2021); Fleming et al. (2020); Peiris-John et al. (2025). Consent without full understanding: attribute constructed at design; no corpus source addresses consent comprehension (post-pilot audit). School-mediated digital access — King-Finau et al. (2021); authorisation persistence, twelve-month re-notification requirement — Ministry of Business, Innovation and Employment (2025). Algorithmic exposure of mixed income — Financial Markets Authority & Reserve Bank of New Zealand (2018); Financial Markets Authority (2024).

Mei-Lin: migrant financial conservatism and privacy — Wang (2018); Asian Family Services & Trace Research (2025); Financial Markets Authority (2024). Mother-translator load across generations — Superu (2015); Ran & Liu (2023). Language and institutional barriers — Hua (2021, November; mechanism evidenced among former refugees and marginalised migrants with English as a second language; transfer to this household is interpretive, post-pilot audit).

Popo: older Chinese immigrant dignity, restraint, privacy — Yeung & Allen (2020); Wang (2018); Asian Family Services (2021); Asian Family Services & Trace Research (2025). Elder anchor role, intergenerational dissonance expressed through care — Superu (2015); Ran & Liu (2023).

Scene: multilingual household structure and base rates — Statistics New Zealand (2020, pp. 124-129, Figure 105; 2021a); Superu (2015); Grimes & White (2019, Table 11).

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### 3 • REFERENCES

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- Hua, J. (2021, November). Digital inclusion user insights — Former refugees and migrants with English as a second language. Department of Internal Affairs. <https://www.digital.govt.nz/dmsdocument/196~digital-inclusion-user-insights-former-refugees-and-migrants-with-english-as-a-second-language/html>
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- Ministry of Business, Innovation and Employment. (2025). Open banking regulations. <https://www.mbie.govt.nz/business-and-employment/business/consumer-data-right/consumer-data-right-policy-design/open-banking>
- Peiris-John, R., Sung, K. H., Sharma, V., Lee, A., Kang, K., Liang, R., Simon-Kumar, R., Ameratunga, S., & Ramalho, R. (2025). Asian youth mental health needs and opportunities: A literature review and summary of research in Aotearoa New Zealand. The University of Auckland.
- Ran, G. J., & Liu, L. S. (2023). Re-constructing reverse family remittances: The case of new Chinese immigrant families in New Zealand. *Journal of Ethnic and Migration Studies*, 49(1), 313-331. <https://doi.org/10.1080/1369183X.2021.1999221>
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- Statistics New Zealand. (2021a). Household income and housing-cost statistics: Year ended June 2021 [Information release]. <https://www.stats.govt.nz/information-releases/household-income-and-housing-cost-statistics-year-ended-june-2021/>

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- Wang, B. (2018). New Chinese migrants in New Zealand: Becoming cosmopolitan? Roots, emotions, and everyday diversity. Routledge.
- Yeung, P., & Allen, J. (2020). Health, quality of life and service needs among older Chinese immigrants: A report prepared for the CNSST Foundation. Health and Ageing Research Team, Massey University.
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#### **4 • LOG**

References converted from Chicago author-date to APA 7. Hedged persona attributions remain in the source document.

Post-pilot source audit. Consent-without-full-understanding is marked constructed: no corpus source addresses consent comprehension; the framework's twelve-month re-authorisation notice is recorded as the adjacent institutional fact. Hua's population warrant is stated in the anchor. Asian Family Services (2021) and Asian Family Services & Trace Research (2025) are successive reports in one survey series; both are now cited plainly. Grimes & White records Asian New Zealanders with the highest internet access of any ethnicity; the scene anchor supports household structure and base rates, not digital exclusion. Statistics New Zealand (2018a) is retired; the scene anchor re-points to Housing in Aotearoa: 2020, Figure 105. A prospective-mechanics note is added to the case position.